



11926 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Yu-Hsuan Teng (PI)	University of Maryland
Prof. Alberto Bolatto (CoI)	University of Maryland
Dr. Adam Leroy (CoI)	The Ohio State University
Dr. Karin Marie Sandstrom (CoI)	University of California - San Diego
Dr. Jiayi Sun (CoI)	University of Kentucky
Dr. Eva Schinnerer (CoI) (ESA Member)	Max Planck Institute for Astronomy
Prof. Sylvain Veilleux (CoI)	University of Maryland
Dr. Rebecca Levy (CoI)	Space Telescope Science Institute
Dr. Thomas Lai (CoI)	California Institute of Technology
Dr. Daniel Dale (CoI)	University of Wyoming
Dr. Torsten Boeker (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Vicente Antonio Villanueva (CoI)	Universidad Diego Portales (UDP)
Jaeyeon Kim (CoI)	Stanford University
Serena Cronin (CoI)	University of Maryland
Keaton Robert Donaghue (CoI)	University of Maryland
Ms. Sara Duval (CoI)	University of Toledo

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Medium Resolution Spectroscopy	(1) M-66

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	4		MIRI Medium Resolution Spectroscopy	(4) M-66-bkgd
	2		MIRI Medium Resolution Spectroscopy	(2) M-100
	5		MIRI Medium Resolution Spectroscopy	(5) M-100-bkgd
	3		MIRI Medium Resolution Spectroscopy	(3) M-95
	6		MIRI Medium Resolution Spectroscopy	(6) M-95-bkgd

ABSTRACT

We propose MIRI-MRS observations of three nearby galaxy centers (NGC 3351, 3627, and 4321) with distinct nuclear environments ranging from pure starburst to active galactic nuclei (AGN). We aim to capture the full H₂ rotational ladder, S(1)–S(8), to directly trace the warm and CO-dark molecular gas at 30-pc resolution. Our sample are selected to ensure the best physical resolution for combined (cold and warm) H₂ analyses, based on robust CO and cold H₂ mass measurements at MIRI-comparable resolutions. We will jointly model the cold and warm molecular phases to map H₂ temperature, mass, and kinematics in the circumnuclear disks (i.e., within 300 pc) of these galaxies. Together with the complete ALMA dataset of CO rotational ladder, our proposed observation will 1) quantify the spatial extent and impact of AGN/starburst heating, 2) measure the full molecular gas reservoir and excitation conditions, and 3) reveal CO-dark gas fractions and feedback imprints in circumnuclear regions. We will establish the first cloud-scale connection between cold and warm molecular gas across a variety of local galactic nuclei, providing key insights into how star formation and AGN activities shape and regulate the molecular ISM.

OBSERVING DESCRIPTION

We request MIRI MRS observations of the nuclear regions (300 pc) in three nearby galaxies: NGC 3351, NGC 3627, and NGC 4321. Each galaxy will be observed in IFU mode to obtain spatially resolved mid-infrared spectra covering the full set of H₂ pure rotational lines from S(1) to S(8). These galaxies are carefully selected based on their nearest distance (to obtain best possible physical resolution) and completeness in ALMA CO datasets at MIRI-comparable resolutions (to ensure the best connection of cold and warm H₂). Our H₂ observations will directly probe the warm (100-2000 K) molecular gas at 0.2-0.4 arcsec (or 10-30 pc) resolution, enabling a detailed study of AGN/starburst heating and the CO-dark molecular gas in the vicinity of local galactic nuclei.

We use a single pointing for the center of each galaxy, with a 4-point dither optimized for extended sources. For each galaxy observation, we arrange a background observation with the same setup but pointing at a nearby emission-free region outside the galaxy. Each galaxy-background pair is set to be observed in a non-interruptible sequence. The exposure time of 40 groups/integration over 1 integration is designed to achieve S/N > 5 for the faintest H₂ 0-0 S(8) line, and we expect to achieve higher S/N of 10-100 for S(1)-S(7) lines. We also set up simultaneous MIRI imaging in filters

JWST Proposal 11926 (Created: Friday, March 13, 2026, 5:05:13PM Eastern Standard Time) - Overview

F560W, F1280W, and F1500W, complementing existing imaging with the F770W, F1000W, F1130W, and F2100W filters. We note that our science goals can be achieved via spectroscopy alone, and thus the simultaneous imaging (which triggers warnings on Imager filter overlap) are simply for added value to the archive. The total science integration time (including background observations) is 2.3 hours for all galaxies, with a total charged time of 8.5 hours.

Proposal 11926 - Targets - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	M-66	RA: 11 20 15.0000 (170.0625000d) Dec: +12 59 29.00 (12.99139d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, Seyfert galaxies]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(2)	M-100	RA: 12 22 54.9300 (185.7288750d) Dec: +15 49 20.30 (15.82231d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, LINER galaxies]</i> <i>Extended=YES</i>	Proper Motion RA: 2.189 mas/yr Proper Motion Dec: 0.26 mas/yr Epoch of Position: 2000	
	(3)	M-95	RA: 10 43 57.7313 (160.9905471d) Dec: +11 42 13.30 (11.70369d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, Starburst galaxies]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(4)	M-66-bkgd	RA: 11 20 36.0000 (170.1500000d) Dec: +12 55 10.00 (12.91944d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, Seyfert galaxies]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(5)	M-100-bkgd	RA: 12 23 16.0000 (185.8166667d) Dec: +15 46 50.00 (15.78056d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, LINER galaxies]</i> <i>Extended=YES</i>	Proper Motion RA: 2.189 mas/yr Proper Motion Dec: 0.26 mas/yr Epoch of Position: 2000	
	(6)	M-95-bkgd	RA: 10 44 10.0000 (161.0416667d) Dec: +11 47 0.00 (11.78333d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Galaxy nuclei, Starburst galaxies]</i> <i>Extended=YES</i>	Epoch of Position: 2000	

Proposal 11926 - Observation 1 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 1												Fri Mar 13 22:05:13 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Observation 1) Warning (Form): Imager Filter overlap.												
	(Observation 1) Warning (Form): Imager Filter overlap.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	M-66	RA: 11 20 15.0000 (170.0625000d) Dec: +12 59 29.00 (12.99139d) Equinox: J2000					Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Galaxy nuclei, Seyfert galaxies] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	

Proposal 11926 - Observation 1 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 1, 4, Non-interruptible

Proposal 11926 - Observation 4 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 4													Fri Mar 13 22:05:13 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Observation 4) Warning (Form): Imager Filter overlap.													
	(Observation 4) Warning (Form): Imager Filter overlap.													
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(4)	M-66-bkgd	RA: 11 20 36.0000 (170.1500000d)					Epoch of Position: 2000						
			Dec: +12 55 10.00 (12.91944d)											
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Acquisition	Category=Galaxy													
	Description=[Galaxy nuclei, Seyfert galaxies]													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					BACKGROUND			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		

Proposal 11926 - Observation 4 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 1, 4, Non-interruptible

Proposal 11926 - Observation 2 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 2													Fri Mar 13 22:05:13 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Observation 2) Warning (Form): Imager Filter overlap.													
	(Observation 2) Warning (Form): Imager Filter overlap.													
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	M-100	RA: 12 22 54.9300 (185.7288750d)				Proper Motion RA: 2.189 mas/yr							
			Dec: +15 49 20.30 (15.82231d)				Proper Motion Dec: 0.26 mas/yr							
			Equinox: J2000				Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
Acquisition	Category=Galaxy													
	Description=[Galaxy nuclei, LINER galaxies]													
	Extended=YES													
Template	#	Target												
	1	NONE												
Dithers	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		

Proposal 11926 - Observation 2 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 2, 5, Non-interruptible

Proposal 11926 - Observation 5 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 5													Fri Mar 13 22:05:13 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Observation 5) Warning (Form): Imager Filter overlap.													
	(Observation 5) Warning (Form): Imager Filter overlap.													
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(5)	M-100-bkgd	RA: 12 23 16.0000 (185.8166667d)					Proper Motion RA: 2.189 mas/yr						
			Dec: +15 46 50.00 (15.78056d)					Proper Motion Dec: 0.26 mas/yr						
			Equinox: J2000					Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Galaxy nuclei, LINER galaxies] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					BACKGROUND			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		

Proposal 11926 - Observation 5 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 2, 5, Non-interruptible

Proposal 11926 - Observation 3 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 3												Fri Mar 13 22:05:13 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Observation 3) Warning (Form): Imager Filter overlap.												
	(Observation 3) Warning (Form): Imager Filter overlap.												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	M-95	RA: 10 43 57.7313 (160.9905471d) Dec: +11 42 13.30 (11.70369d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Galaxy nuclei, Starburst galaxies] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006	
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	

Proposal 11926 - Observation 3 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 3, 6, Non-interruptible

Proposal 11926 - Observation 6 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei

Observation	Proposal 11926, Observation 6												Fri Mar 13 22:05:13 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Observation 6) Warning (Form): Imager Filter overlap.													
	(Observation 6) Warning (Form): Imager Filter overlap.													
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(6)	M-95-bkgd	RA: 10 44 10.0000 (161.0416667d)				Epoch of Position: 2000							
			Dec: +11 47 0.00 (11.78333d)											
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Acquisition	Category=Galaxy													
	Description=[Galaxy nuclei, Starburst galaxies]													
	Extended=YES													
Template	#	Target												
	1	NONE												
Dithers	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Spectral Elements	#	Dither Type				Optimized For				Direction				
	1	4-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2		IMAGER	F1280W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3		IMAGER	F1500W	FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006		
	3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006		

Proposal 11926 - Observation 6 - A Complete Accounting of Molecular Gas and AGN/Starburst Heating in Local Galactic Nuclei	
Special Requirements	Sequence Observations 3, 6, Non-interruptible